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Change light vehicle tyre

Document Type	ByrneCut Group Safe Work Instruction (SWI)
Document Number	MAIN-SWI-017
Description	Change light vehicle tyre

OVERVIEW

Purpose

To establish the process for the safe change out of a light vehicle tyre outside the workshop environment to ensure that all such processes are executed in a safe and efficient manner and that equipment used functions correctly and in accordance with their design intention.

Scope

SWI Standard Scope

This safe work instruction applies to all burn cut employees and contractors. All employees are required to act in accordance with safe standards established in the process. Only persons approved and competent shall carry out this task.

Prior to starting the task

Be competent in the following: Mate Pro 1; Burn cup group maintenance induction; B-CORT M 0018 Mobile equipment operational safety; B-CORT M 0047 Isolation main manual; B-CORT M 0003 Light vehicle operation. Refer to M-P-N 267 (light vehicle will not over torque). Before any repairs commence, have clear instructions from your supervisor or leading hand. If no suitable procedures exist, complete a JHA before commencing work.

PPE	EQUIPMENT	ENVIRONMENT
• Safety Glasses • Steel Capped Boots • Trousers and long-sleeved shirts rolled down. • Hard Hat if applicable	• wheel chocks	• Barricaded Area • Machine Isolated

ITEM	STEP	PRINCIPAL HAZARDS
Preparation	• Are you authorised to perform the task? • Do you have the correct tickets? • Ensure you are familiar with all aspects of the scope of work to be carried out. • If unsure, ask your supervisor. • Light vehicle must be isolated as per B-CORT M0047 Isolation main manual. • Where required, only engineered and rated stands, cribbing blocks and locking devices must be used to ensure the item cannot fall.	07 - Fitness for Work 09 - Human Factors 02 - Crush / Caught by Equipment



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ITEM	STEP	PRINCIPAL HAZARDS
	Park vehicle off main decline on flat level ground; move if safe to do so. Select first or reverse gear and apply handbrake.	
General safety rules	- Never work under a supported or suspended load. - Never allow any part of your body under any part of the vehicle held up by a jack only. - Identify and avoid pinch points. - Wear gloves for sharp edges. - Be careful around equipment ensuring not to trip or slip. - Clean up fluid spills straight away and roll up hoses after use. - Ensure adequate fluid intake and frequent rest periods; observe team members for signs of dehydration and fatigue. - Ensure machine barricades installed prior to commencement of tasks if applicable. - Wheel chocks to be installed prior to commencement of work.	02 - Crush / Caught by Equipment 14 - Rotating Equipment 09 - Human Factors 07 - Fitness for Work
Work area preparation	Communicate with other road users via the radio and take extra precautions such as putting warning signs, traffic triangles and/or a spotter in high traffic areas.	15 - Single Mobile Plant Interactions
Task steps	- Remove spare wheel from live vehicle ensuring there is no chance that the tyre will run away or roll down the decline. - When removing spare tyre from a utility tray or spare tyre holder ensure that the tyre does not land on any part of your body or be in a situation where it can bounce and injure you; get assistance and lift the tyre safely to the ground. Use safe manual handling (keep back straight). - Roll or slide tyre rather than lifting full weight. - Loosen the wheel nuts on the wheel that is being changed (just crack them). - Position the jack at the correct jacking point on level ground only. There are different types of jacks that can be used. - Ensure all personnel understand you're about to raise the vehicle and are standing well clear. - Where possible position the spare tyre under the chassis differential so that if the vehicle slides off the jack it will come to rest on the spare tyre. - Raise the vehicle so that the wheel is clear of the ground and high enough to put the new wheel on. - Check your chocks are still in a suitable position and have not moved in the jacking process.	02 - Crush / Caught by Equipment 07 - Fitness for Work 16 - Tyres and Rims 15 - Single Mobile Plant Interactions 09 - Human Factors



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ITEM	STEP	PRINCIPAL HAZARDS
	• Never allow any part of your body under any part of the vehicle held up by a jack only. • Remove the wheel nuts and place them in a safe location out of the way of dirt, side steps or tray. • Remove the wheel carefully; be gentle as sharp forceful movements can cause the machine to slip off the jack. • Ensure the hub and spare wheel mating faces are clean with no build up of dirt, rust etc. that can lead to the wheel coming loose later. • Inspect the taper on the nuts and the nut seat on the rim for distortion; check wheel stud threads for signs of over tension. • Fit the new wheel onto the hub using correct manual handling techniques. • Reinstall the wheel nuts tapered end inwards. • Hand tighten all wheel nuts and then tighten them in the correct pattern. • Lower the vehicle completely. • Tension the wheel nuts with a wheel brace or torque wrench if available. If using a torque wrench ensure there is movement during the final torque; if no movement the wheel nuts may be over tightened. • Install wheel nut indicators where fitted. Use specified torque values (e.g., Land Cruiser 209 N·m; Hilux 105 N·m; 2WD Hiace 100 N·m; 4WD Hiace 180 N·m). • Secure the old flat tyre back onto its original location and place outer service tag on the wheel. Get assistance to lift the tyre if needed. • Place a report or information tag onto the ignition key or grab handle with the date, time and kilometres the wheel was changed so that the wheel nuts can be re-tensioned. Ensure retork book has been filled in. If red rim has been fitted the light vehicle is to be taken to maintenance for replacement. • Ensure all tooling and jack is returned securely to the light vehicle. • Ensure flat tyre is replaced on vehicle. • Final check: ensure wheel nuts are re-tensioned and all tools are stowed.	



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SIGN OFF SECTION - ACKNOWLEDGEMENT OF READING AND UNDERSTANDING

Name		Signature	
Date			